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图纸文件目录（首页）

CATALOG OF DRAWING DOCUMENTS (HEAD PAGE)

拉合尔±660kV换流站 工程 施工图 阶段
±660kV Lahore Converter Station PROJECT Detail Design DESIGN STAGE

消防 专业 第 02 卷 第 11 册
Fire Fighting SPECIALITY VOLUME BOOK

卷册名称 巴方宿舍给排水管道布置图（第一次修改）
Volume name Dormitory Building for Pakistani Water Supply & Drainage Plane Arrangement Drawing (V1.0)

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审 核 REVIEWED BY	
设计/勘测 DESIGNED BY/ SURVEYED BY	
图 纸 8 张 DRAWING NOS.	打印件 1 本 PRINTED COPIES NOS.

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注:

1. 总则：

1. General:

1.1. 本图标高单位以米计，管径及尺寸单位以毫米计，标高采用相对高程系统。图中室内±0.00米标高相当于EGM2008高程系统标高193.85m。室内外高差为0.45m。

1.1 The unit of elevation is m and that of all dimensions（except otherwise specified）is mm. Relative elevationsystem is applied: 0.00m elevation equals 193.85m EGM2008 elevation. Elevation difference between building ground elevation and plant ground elevation is 0.45m. Architecture AB coordinate system is used. And for detail information of coordinate system, please refer to plant general layout drawing.

1.2. 本图图中凡套用《国家建筑标准设计图库》中的部分，应按标准图集的要求备料施工，不另出图。

1.2 For those designed in accordance with "National Building Standard Design Atlas", constructions & installation shall be in strict accordance with the Atlas without any other dwg.

1.3. 工程中所采用的设备、器具与材料，其品种、规格与技术性能等，均应符合国家的现行标准与工程的设计要求；并应具有产品质量合格证，严禁采用不合格产品。

1.3 The specifications and performance parameters of all equipment, applicants and materials shall be in accordance with relevant codes and standards with quality certifications.

1.4 本工程场区生活水管网水源为深井泵处理后压力水，水压为0.4MPa，详见本工程30-B003701S-S0201卷册。

1.4 The water supply of water supply pipe network in this project station is the pressure water from the deep well pump, and the water pressure is 0.4MPa，see the detail on the volume:30-B003701S-S0201.

1.5. 本工程的施工与验收，应按照如下规范与该工程的设计要求进行：

1.5 Construction, installation and acceptance shall be in accordance with following codes and standards as well as design requirements:

（1）《建筑给水排水设计规范》GB 50015-2009；

（1）《Code for design of building water supply and drainage》GB 50015-2009；

（2）《建筑给水排水及采暖工程施工质量验收规范》GB 50242-2002；

（2）《Code for acceptance of construction quality of Water supply drainage and heating works》GB 50242-2002；

（3）《给水排水管道工程施工及验收规范》GB 50268-2008；

（3）《Code for construction and acceptance of water and sewerage pipeline works》GB 50268-2008；

（4）《建筑给水塑料管道工程技术规程》CJJ/T98-2014；

（4）《Technical specification for plastic pipeline engineering of building water supply》CJJ/T98-2014；

（5）《建筑排水柔性接口铸铁管管道工程技术规程》CECS 168-2004；

（5）《Technical specification for flexible joint cast iron pipe system of building drainage》CECS 168-2004；

（6）《排水用柔性接口铸铁管、管件及附件》GB/T 12772-2008；

（6）《Cast iron pipes: fittings and accessories with flexible joint sewerage for drainage》GB/T 12772-2008；

（7）《建筑防火封堵应用技术规程》CECS 154：2003

（7）《Technical specification for application of fire stopping in buildings》CECS 154：2003

（8）《建筑机电工程抗震设计规范》GB 50981-2014；

（8）《Code for seismic design of mechanical and electrical equipment》GB 50981-2014；

（9）《给水排水标准图集》。

（9）《Standard collective drawing for water supply and drainage》。

1.6. 管道的安装及施工应有完整的施工质量验收记录。

1.6 A complete piping installation and construction quality acceptance record shall be well performed and kept.

2. 管道材料

2. Pipe material

2.1. 生活给水管（冷水管）：采用建筑给水用PP-R（PN1.0）S5级管道及相应的配套管件，管径以dn表示。生活给水管（热水管）采用建筑给水用PP-R（PN1.0）S3.2级热水管道，设计温度60°C，管径以dn表示。管材标准应满足《冷热水用聚丙烯管道系统 第1部分：总则》GB/T 18742.1-2002，《冷热水用聚丙烯管道系统 第2部分：管材》GB/T 18742.2-2002，《冷热水用聚丙烯管道系统 第3部分：管件》GB/T 18742.3-2002。

2.1. Potable water supply pipe (cold water): PP-R (PN1.0) S5-level pipeline and the corresponding supporting pipe, and the pipe diameter is in the name of dn. Potable water supply pipe (hot water): PP-R (PN1.0) S3.2 grade hot water pipe, and the design temperature is 60°C, and the diameter is in the name of dn.. Standard for piping should meet the "Polypropylene piping system for hot and cold water" Part 1: General "GB / T 18742.1-2002," Polypropylene piping system for hot and cold water Part 2: Tubing "GB / T 18742.2-2002," Cold Polypropylene piping systems for hot water - Part 3: Fittings GB / T 18742.3-2002.

2.2. 排水管：室内排水管采用PVC-U排水塑料管道以及相应的配套管件，管材标准为《建筑排水用硬聚氯乙烯（PVC-U）管材》GB 5836.1-2006，管径以de表示。首层及室外排水管采用衬塑钢管以及相应的配套管件，管材标准为《钢塑复合管》GB/T 28897-2012和《建筑给水钢塑复合管管道工程技术规程》CECS125:2001，管径以DN表示。

2.2. Sanitary sewage piping: indoor drain pipe adopts PVC-U drainage plastic pipe and corresponding supporting fittings, and the pipe standards shall be meet that "Unplasticized poly (vinyl chloride) (PVC-U) pipes for soil and waste discharge inside buildings" GB 5836.1-2006. The pipe diameter is in name of de. The first floor and outdoor drains are lined with plastic steel tubes and corresponding supporting fittings, and the pipe standards shall be meet that, "Steel pipes of complex plastic" GB/T 28897-2012 and "Technical specification for steel-plastic complex pipeline engineering of water supply in building" CECS125:2001. The pipe diameter is in name of DN.

3. 管道加工

3. Pipeline processing

3.1. 裁截塑料管宜用钢锯切割或其它机械切割，不允许用敲击的方法裁截。

3.1 Steel saws or other mechanic devices shall be used to cut plastic piping. No manual cuttings, such as knocking and tearing, shall be permitted.

3.2. 管子切口要平整干净，并不得有裂纹与重皮。

3.2. Piping notch shall be clean and neat without any cracks.

3.3. 衬塑钢管应采用专用切管器垂直于管轴线截管，不得采用火焰切割、电焊切割。

3.3. The lining plastic pipe should be used special pipe cutter to cut tube perpendicular to the tube, not to use flame cutting, welding cutting.

4. 管道连接

4. Pipeline connection

4.1. 给水管和排水管的连接应严格按照制造厂提供的连接工艺及方法进行。

4.1. The connection of the water supply pipe and the drain pipe shall be carried out in strict accordance with the connection procedure and method provided by the manufacturer.

4.2. PP-R管和管灵配件之间应采用热熔连接，并应采用专用热熔工具，PP-R管与金属管件连接，应采用带金属嵌件的聚丙烯管件为过渡，该管件与PP-R管采用热熔连接，与金属管件或卫生器具金属配件采用丝扣连接。室内PP-R管与室外生活给水钢管管材过渡在管道开到地上。室外生活给水管具体要求参照站区生活给水管卷册说明。

4.2. PP-R pipe and pipe and fittings should be jointed by the hot melt way, and should use special hot-melt tools, the connection between PP-R pipe and metal pipe fittings should use metal insert of polypropylene pipe for the transition, the pipe And PP-R pipe with hot-melt connection, and metal fittings or sanitary fittings with metal accessories threaded connection. Indoor PP-R pipe with outdoor living water supply steel pipe transition in the pipeline to the ground. The specific requirements of the water supply pipe refer to the volume of yard domestic water pipe network.

4.3. 室内PVC-U排水管采用粘接连接，PVC-U排水管与室外混凝土管根据厂家提供的连接方法及工艺连接，在室外过度。

4.3. Indoor PVC-U pipes connected by adhesive,PVC-U drainpipe and outdoor concrete pipe are connected by the connection method and process provided by the manufacturer, and excessive outdoors.

5. 管道安装

5. Pipeline installation

5.1. 水平管道的设计标高：给水管是指管中心，排水管是指管内底。排水管坡度除标注外宜向横干管及出户管de50：i≥0.025；de75：i≥0.015；de110：i≥0.012；de160：i≥0.010；室外排水干管D200，i≥0.005，D300，i≥0.004。

5.1 Horizontal piping elevation: that of supply piping is piping central elevation; and that of drainage piping is internal bottom elevation. Drainage piping slope:de50：i≥0.025；de75：i≥0.015；de110：i≥0.012；de160：i≥0.010；D200，i≥0.005，D300，i≥0.004。

5.2. 在安装管道之前，必须清除管内杂物以及承口内部与插口外部的杂质。

5.2. Before piping installation, the inside of piping and sockets area shall be well cleaned.

5.3. 给排水管道布置可根据现场实际情况调整，管道的固定管卡和吊架的具体位置由安装时在现场决定。

5.3. The layout of the water supply and drainage pipes can be adjusted according to the actual situation of the site. The specific location of the pipe and hanger shall be determined by site operation.

5.4. 给水管道室内尽可能暗装。明装管道时，管道的连接处应设置在墙壁和梁板外面，明装的横管不应有明显的弯曲。

5.4. The water supply piping installation shall be concealed as much as possible. For exposed piping, piping connection shall be set in the outside of wall or floor and no visible severe bending shall be allowed.

5.5. PP-R管支吊架安装说明详见《给水排水标准图集》11S405。

5.5. The installation instructions of PP - R pipe support bracket are detailed in the water supply and drainage standard atlas, 11S405.

5.6. 给水管道穿越墙体、室内地面及楼面应安装套管并填灰，具体做法见图集11S405。

5.6. The water supply pipe through the wall, indoor floor and floor should be installed with casing and filling in. The specific method refers to atlas 11S405.

5.7. 严禁生活饮用水管道与大便器（槽）、小便斗（槽）采用非专用冲洗阀直接连接冲洗。

5.7. It is strictly forbidden to use the non-special flush valve to flush the water pipe and urinal (tank) and urinal directly.

5.8. 在非饮用水管道上接出水嘴或取水短管时，应采取防止误饮误用的措施。

5.8. Measures to prevent accidental drinking misuse should be taken when the non-drinking water pipes are connected to the water outlet or the short pipes.

5.9. 给水管网末端和最高点以及有明显起伏积聚空气的管段应设置自动排气阀。

5.9.The automatic exhaust valve shall be installed at the end and the highest point of the water supply pipe network and the pipe section with obvious fluctuations.

5.10. 给水管道不得敷设在烟道、风道、电梯井内、排水沟内，不得穿过大便槽和小便槽，且立管离大便槽、小便槽端部不得小于0.5m。

5.10. The water supply pipeline shall not be laid in the flue, duct, elevator shaft and drainage ditch. It is not allowed to pass through the trough and urinal, and the riser is not less than 0.5 m from the trough and the end of the urinal.

5.11. 给水管道不宜穿越伸缩缝、沉降缝、变形缝，如必须穿越时，应设置补偿管道伸缩和剪切变形的装置。

5.11. The water supply pipe should not travel through expansion joints, settlement joints and deformation joints. If you must cross, you should set up a device that compensates for pipe expansion and shear deformation.

5.12. 密闭的水加热器或用水设备的进水管上应设置止回阀。

5.12. Check valves shall be installed on the inlet pipe of closed water heater or water equipment.

5.13. 排水管支吊架安装说明及支吊架详图见《给水排水标准图集》11S409。

5.13. Installation instructions of drainage pipes and a detailed drawing of the support hanger refer to water supply and drainage standard atlas, 11S409.

5.14. 排水管道穿地下室外墙、砖墙基础、楼面、屋面及埋地安装图详见《给水排水标准图集》13S409 34、35。

5.14. The drainage pipe is in the basement wall, brick wall foundation, floor, roof and buried installation drawings. Refer to the water supply and drainage standard atlas, 13S409 34, 35.

5.15. 排水立管与排出管端部的连接，宜采用两个45°弯头、弯曲半径不小于4倍管径的90°弯头。

5.15. Two 45° elbows or 90° elbows of more than 4 times piping diameter turning radius are recommended to connect drainage vertical piping and outlet horizontal piping.

5.16. 卫生器具排水管与排水横支管垂直连接，宜采用90°斜三通。

5.16. the 90° inclined tee is recommended to connect sanitary ware cross drainage and drainage branch pipe when they are vertical connected.

5.17. 排水管道的横管与立管连接，宜采用45°斜三通或45°斜四通和顺水三通或顺水四通。

5.17. 45° inclined tee or 45° inclined cross and well tee or well cross are recommended to connect horizontal tube and vertical tube drainage pipe.

5.18. 排水立管应避免在轴线偏置；当受条件限制时，宜用乙字管或两个45°弯头连接。

5.18. Drain riser should be avoided on axis bias; if conditions is restricted, pipe offset or two 45° elbow connection should be recommended.

5.19. 当排水支管、排水立管接入横干管时，应在横干管管顶或其两侧45°范围内采用45°斜三通接入。

5.19. When the drainage pipes, drainage risers connect the main horizontal pipes shall use 45° inclined tees to connect at the top of the horizontal main tube or on both sides of 45°.

5.20. 排水立管上的检查口安装高度距地面为1.0米，并应高于该层卫生器具上边缘0.15m。

5.20. The height of the inspection port on the drainage riser is 1.0 meters from the ground, and should be 0.15m above the edge of the sanitary appliance.

5.21. 衬塑钢管与塑料排水管的连接，参见国标图集10S406《建筑排水塑料管道安装》、04S301《建筑排水设备附件选用安装》。

5.21. Connection of lining plastic pipes and plastic drain pipes, refers to the national standard atlas 10S406, "construction drainage plastic pipe installation", and 04S301 "installation of the accessories for building drainage equipment".

5.22. 当建筑铸铁排水管穿越楼层、防火墙、管道井井壁时，应根据建筑物性质、管径和设置条件以及穿越部位防火等级等要求设置阻火装置。当管道穿过防火墙时，管道与防火墙之间的缝隙应采用防火材料堵塞。当直径大于等于32mm的管道穿过防火墙时，除填塞防火材料外，还应采取阻火措施。措施包括采用阻火圈或阻火带并辅以防火泥或防火密封胶的有机堵料等封堵。本工程中，涉及到管道穿越砖砌机房隔墙及楼板，应采用防火泥进行封堵。

5.22. When cast iron drain pipes across the floor, firewall, pipeline wall, it shall be according to the type of the building, pipe diameter, and set conditions as well as the parts through which the pipes across requires. site fire rating set fire equipment. When the pipe passes through the firewall, the gap between the pipe and the firewall should be blocked with fireproof material. When the diameter of the pipe is greater than or equal to 32mm, the fire resistance should be adopted in addition to the fireproof material. The measures include the use of a fire resistance ring or a fire retarding belt and a combination of organic blocks, such as fireproof mud or fireproof sealants, etc. In this project, the pipe is involved in the wall and the floor of the smoking room.

5.23. 排水横管上的清扫口宜设置在楼板或地坪上，且与地面相平；排水横管起点的清扫口与其端部相垂直的墙面的距离不得小于0.2m。排水管道设置的清扫口，其材质应为铜质。

5.23. The clearance of the drainage tube shall be set on the floor or on the floor of the floor, and flat with the ground; The distance between the opening of the discharge line and the vertical wall of the end shall not be less than 0.2m. Drainage pipe set for cleaning, its material should be copper.

5.24. 排水管起点设置堵头代替清扫口时，堵头与墙面应有不小于0.4m的距离，可利用带清扫口的弯头配件代替清扫口。

5.24. When the end cap replaces the cleanout at the starting point of the drainage pipe, the end cap and the wall should not be less than 0.4m in distance, and the elbow fittings with cleanout should be used instead of the cleanout.

5.25. 通气管若高出屋面（屋顶有隔热层时，应从隔热层板面算起），则离出屋面的部分不得小于0.3m，且应大于屋顶覆土厚度并叠加最大积雪厚度，通气管顶端应装设风帽或网罩。

5.25. If a vent pipe above the roof (if the roof has an insulating layer, the height shall be calculate from the insulating layer), the height higher than the roof shall not be less than 0.3m, and shall be greater than soil thickness and the maximum snow thickness, and the cap and net cover shall be installed for the vent pipe.

5.26. 在通气管口周围4m以内有门窗时，通气管口应高出窗顶0.6m或引向无门窗一侧。

5.26. When there are doors and windows within 4 meters of the air pipe, the vent should be 0.6m above the window or to the side of the windowless side.

5.27. 排水管道不得穿过伸缩缝、沉降缝、变形缝、烟道和风道；当排水管道必须穿过沉降缝、伸缩缝和变形缝时，应采取相应技术措施。

5.27. Drainage pipes shall not pass through expansion joints, settlement joints, deformation joints, flue and duct; The corresponding technical measures should be adopted when the drainage pipe must go through the settlement joint, expansion joint and deformation joint.

5.28. 浴室、食堂地漏采用带网罩有水封型，且浴室地漏引用毛发聚集器；热水设备间排水沟采用直通式地漏。

5.28. In the bathroom, the floor of the dining room shall be water sealed with net basket, and the hair collector shall be used in the bathroom floor. The drain in the hot water equipment is used in a straight - way floor drain.

5.29. 热水管及室外外露管道需外包40mm厚的离心玻璃棉及0.5mm厚的铝箔保温。

5.29. Hot water pipes and outdoor exposed pipes need to be outsourced 40mm thick centrifugal glass wool and 0.5mm thick aluminum foil insulation.

6. 管道系统的试验

6. Test of piping system

6.1. 室内给水管道应进行水压试验，生活水管工作压力约0.40MPa（站区生活水管网），试验压力0.90MPa。管道所采取的连接方式需大于可承受的试验压力。

6.1. Hydraulic pressure test shall be performed for indoor water supply piping. The working pressure is about 0.40 MPa (the factory life water pipe network), and the test pressure is 0.90 MPa. The connection mode of the pipe shall be greater than the test pressure.

6.2. 排水管道应进行灌水试验。

6.2. Drainage pipes shall be tested for irrigation.

6.3. 水压试验或灌水试验应作试验记录。

6.3. Test records shall be made for water pressure test or irrigation test.

6.4. 饮用水管水压试验完毕后必须进行冲洗和消毒。

6.4. Rinse and disinfection must be carried out after the drinking water pressure test is completed.

6.5. 卫生器具给水配件承受的最大工作压力不得大于0.6MPa。

6.5. The maximum working pressure for water fittings of sanitary appliances shall not be greater than 0.6MPa.

7. 卫生器具及附件的安装

7. Installation of sanitary appliances and accessories

7.1 卫生器具给水配件安装时，镀铬零件必须完好，连接口必须严密，无渗漏现象。

7.1 when the plumbing fittings are being installed, the chrome plated parts must be in good condition and the connection must be tight and no leakage.

7.2. 卫生器具支、托架的安装必须平整、牢固，支架与卫生器具接触应紧密。

7.2. The installation of sanitary appliances and brackets shall be flat and firm, and the supports shall be in close contact with hygienic instruments.

7.3. 卫生设备的安装详图见《给水排水标准图集》09S304和04S301，宜采用膨胀螺栓或用预埋支架进行固定，如采用木螺丝固定的，应采用嵌入墙内经防腐处理后的楔形木砖，且木砖应深入净墙面100mm。

7.3. Sanity wares installation details refer to atlas of water supply and drainage standard 09 s304 and 04 s301. Embedded bolts or plug bolts shall be used to install sanitary wares. If any wooden bolts are used, embedded wooden bricks anti-corrosion treatment shall be performed, which shall be 100mm inside the wall surface.

7.4. 卫生器具的安装高度和卫生器具给排水配件的安装尺寸应符合《建筑给水排水及采暖工程施工质量验收规范》的要求。

7.4. Installation dimensions of sanitary wares shall be in accordance with "Code for Acceptance of Construction quality of Water Supply Drainage and heating Works"（GB 50242-2002）.

7.5. 卫生设备上下水留孔位置应根据产品要求确定，故设备到达现场后复核尺寸后再进行留孔。安装时应注意和土建配合，安装后用燃材料密封好。

7.5. The position of the water retention hole of the sanitary equipment shall be determined according to the requirements of the product, so the equipment shall be rechecked after the site is returned to the site. The installation should pay attention to the construction of the installation, after installation, sealed with flame retardant material.

7.6. 所有地漏及卫生器具均应设存水弯或水封装置，水封深度不得小于50mm。采用普通带水封地漏，地漏顶面低于安装地面2cm。

7.6. All floor leakage and sanitary appliances shall be equipped with water or water sealing devices, and the water seal depth shall not be less than 50mm. The surface of floor should be 2cm above the top of the floor drain .

7.7. 卫生器具和用水设备、构筑物等的生活饮用水管配水件出水口不得被任何液体或杂质所淹没；出水口高出承接用水容器溢流边缘的最小空气间隙，不得小于出水口直径的2.5倍。

7.7. The water outlet of drinking water pipes with water pipes and water fittings shall not be submerged by any liquid or impurities; The minimum air gap between the outlet and the overflow edge of the water container shall not be less than 2.5 times the diameter of the outlet.

7.8. 卫生器具排水管段上不得重复设置水封。

7.8. Water seals shall not be repeated on the sanitary fittings.

8. 抗震设计

8. Aseismatic design

8.1. 根据本工程岩土勘察，拟建场地的地基液化等级均判为中等液化，因此设计时考虑地基土的地震液化影响；污水检查井采用钢筋混凝土检查井，排水管道采用铸铁排水管等。

8.1. According to the geotechnical data of this project，that the proposed site were jailed for medium liquefied foundation liquefaction grade, the effects of earthquake liquefaction of foundation soil are considered in the design: sewage well using reinforced concrete inspection well, drainage pipe using cast iron pipes, etc.

8.2. 本工程抗震设防烈度为7度，机电设备及管道的抗震措施应符合规范《建筑机电工程抗震设计规范》GB50981-2014。

8.2. The seismic fortification intensity of this project is 7 degrees, and the anti-seismic measures of mechanical and electrical equipment and pipelines should conform to the specification for seismic design of building mechanical and electrical engineering (GB50981-2014).

8.3. 管道穿过内墙或楼板时，应设置套管；套管与管道间的缝隙，应采用柔性防火材料封堵。

8.3. When the pipe runs through the wall or floor, the casing shall be set: the gap between casing and pipe should be sealed with flexible fireproof material.

8.4. 管道支架需符合《建筑机电工程抗震设计规范》GB50981-2014中的相关规定。

8.4. The pipe supports shall conform to the relevant provisions in gb50981-2014 of the design specification for seismic design of mechanical and electrical engineering.

8.5. 运行时不产生振动的给水水箱、太阳能设备等设备、设施应与主体结构牢同连接，与其连接的管道应采用金属管道。

8.5. The equipment and facilities such as water tank and solar equipment that do not produce vibration during operation shall be connected with the main structure of the main structure, and the pipe connected to it shall be in metal pipe.

9. 建筑节能

9. Building energy saving

9.1. 所有水泵、设备等采用低噪音高效率环保型产品。

9.1. All water pumps, equipments etc should use low noise high efficiency environmental friendly products.

9.2. 全部给水配件和卫生洁具均采用节水型产品，并应符合国家现行标准《节水型生活用水器具》的有关规定，同时应符合《民用建筑节水设计标准》GB50555-2010中第6.1节的相关要求；卫生洁具给水及排水五金配件应采用与卫生洁具配套的节水型配件，不得采用淘汰产品。

9.2. All water fittings and sanitary ware shall use water-saving products, and shall be consistent with the relevant provisions of the current national standards of the water-saving appliances. At the same time it should comply with the standards for design of civil construction of water-saving GB50555-2010 in section 6.1 of the related requirements; Sanitary ware supply and drainage hardware fittings should be used with the water-saving fittings with sanitary fittings, and the products eliminated should not be used.

9.3. 给水水嘴应采用陶瓷芯等密封性能良好、能预制出流率并经国家有关质量检测部门检测合格的节水水嘴。

9.3. The water supply nozzle should adopt a good sealing performance such as ceramic core, which can limit the flow rate and test qualified water-saving water nozzle by the national quality inspection department.

9.4. 坐便器冲洗水箱一次冲洗容积不应大于5升，尽可能采用3.5/5升两档冲洗水箱。

9.4. The flushing tank of the toilet flushing tank should not be larger than 5 liters, as far as possible 3.5/5 liters of flushing water tank.

9.5. 给水管道均选用摩阻系数较小的新型塑料给水管材和复合给水材料，选用密封性能好的阀门、设备并保证所有设备、管材及附件的工作压力不大于产品标注的允许工作压力。

9.5. Water supply pipeline smaller friction coefficient was used for the new plastic water supply pipe and composite materials. Both of valve and equipments with good sealing performance should be used. and it shall be ensured that all equipment, pipes and fittings of working pressure is greater than the allowable working pressure of product labeling.

10. 其它说明

10. Other notes

10.1. 本册材料表中仅包括主要管道卫生设备器材，零星配件材料按现场实际需要配备。

10.1. Only major piping, valves and sanitary wares are listed in the list of materials of this drawing, and other fittings shall be procured according to actual installation requirements.

10.2. 室内上下水管道与室外上下水管道接口位置以本册图为准。

10.2. The location of the upstream and upstream pipelines of the indoor and upper water pipes shall be subject to the manual.

10.3. 室外生活污水检查井采用钢筋混凝土污水井，施工详图见给水排水标准图集02S515。当检查井位于非道车地坪上时，井顶标高高出周围地坪5cm，当检查井位于道车地坪上时，井顶与周围地坪齐平。

10.3. The reinforced concrete sewage wells shall be used for the outdoor living sewage inspection wells ,which refers to the detailed drawings of the water supply and drainage standards 02S515. When the inspection well is located on the non-open floor, the top elevation of the well is 5cm higher than the surrounding floor. When the inspection well is located on the open floor , the top of the well is flush with the surrounding floor.

10.4. 金属管道的防腐应符合现行的国家有关卫生标准的要求，具体参照站区生活给水管卷册说明。

10.4. The embalming of metal pipes should conform to the current state requirements for hygienic standards, specifically referring to the description of water pipe volumes in the plant life.

11. 图例：

11. Legends:

图 例 （Legends）

序号 No.	卫生设备名称 Sanitary Ware	下水管道图例 Drainage Legend	标准图集名称 Reference of ARCHLJB
1	洗涤盆 Sink		给水排水标准图集09S304 ARCHLJB 09S304
2	洗脸盆 Lavatory		给水排水标准图集09S304 ARCHLJB 09S304
3	大便器 Toilet		给水排水标准图集09S304 ARCHLJB 09S304
4	小便器 Urinal		给水排水标准图集09S304/101 ARCHLJB 09S304/101
5	地漏 Floor Drain		给水排水标准图集04S301/30 ARCHLJB 04S301/30
6	清扫口 Clean-out opening		给水排水标准图集04S301/13 ARCHLJB 04S301/13
7	水表 Water Meter		给水排水标准图集01SS105/8 ARCHLJB 01SS105/8

检查口 Inspection Opening	伸顶通气管 Roof Vent	波纹补偿器 Ripple Compensator
截止阀 Stop Valve	异径管 Reducer	存水弯 Water Trap
给水管 Supply Piping	排水管 Drainage Piping	自动排气阀 Automatic Exhaust Valve
减压阀 Pressure relief valve	带金属管件的聚丙烯接头 Polypropylene connector with metal tubing	

（附表一）给排水管外径 壁厚与公称直径对照关系

（Attached Table I）Contrasting relations between and among external diameter and wall thickness and nominal diameter.

冷水管（S5级）PP-R管外径 dn（mm） External diameter of PP-R pipes(S5) for cold water dn（mm）	20	25	32	40	50	63	75	90
公称直径 DN（mm） Nominal diameter DN（mm）	15	20	25	32	40	50	65	80
公称壁厚 （mm） Nominal wall thickness （mm）	–	2.3	2.9	3.7	4.6	5.8	6.8	8.2
热水管（S3.2级）PP-R管外径 dn（mm） External diameter of PP-R pipes(S3.2) for hot water dn（mm）	20	25	32	40	50	63	75	90
公称直径 DN（mm） Nominal diameter DN（mm）	15	20	25	32	40	50	65	80
公称壁厚 （mm） Nominal wall thickness （mm）	2.8	3.5	4.4	5.5	6.9	8.6	10.3	12.3
PVC-U管外径 de（mm） External diameter of PVC-U drain pipe de（mm）	50	75	110	160				
公称直径 DN（mm） Nominal diameter DN（mm）	50	75	100	150				
混凝土管 公称直径 DN（mm） Nominal diameter of concrete pipe DN（mm）	100	300						
混凝土管壁厚 （mm） Thickness of concrete pipe （mm）	25	30						

Reference Drawings

Drwg. No.	Title

Revisions

Name	Date	Notes

巴基斯坦默蒂亚里一拉合尔±660kV直流输电工程

±660kV HVDC Project from Matiari to Lahore in Pakistan

NATIONAL TRANSMISSION & DESPATCH COMPANY (NTDC) LIMITED PAKISTAN

CONTRACTOR 国家电网 State Grid 中国电力技术装备有限公司 CHINA ELECTRIC POWER EQUIPMENT AND TECHNOLOGY CO.,LTD.

SUBCONTRACTOR East China Electric Power Design Institute Co.,Ltd.

拉合尔±660kV换流站

±660kV Lahore Converter Station

设计阶段 PHASE：施工图 Detail Design

M L H V D C - L C S - E C I - D D - W F - 0 2 1 1 - 0 0 1 A

	姓名 Name	日期 Date
设计 Designed	朱星跃	2018.01.11
校核 Checked	张科杰	2018.01.11
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巴方宿舍

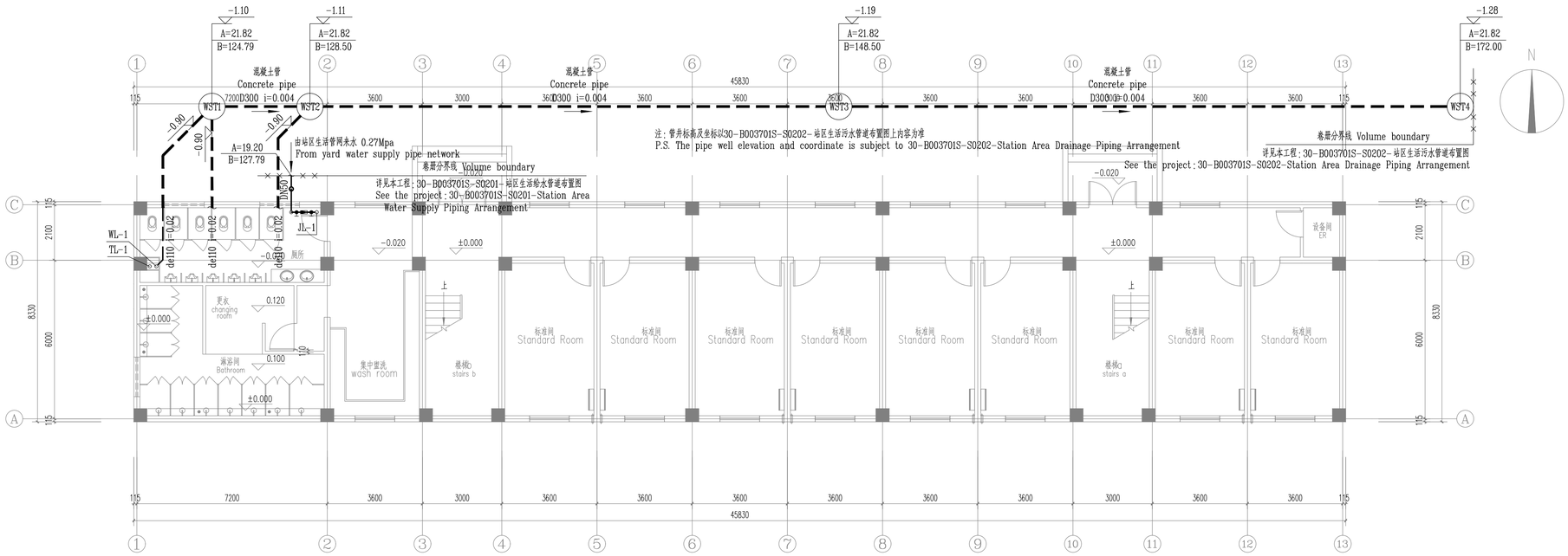
Dormitory Building for Pakistani

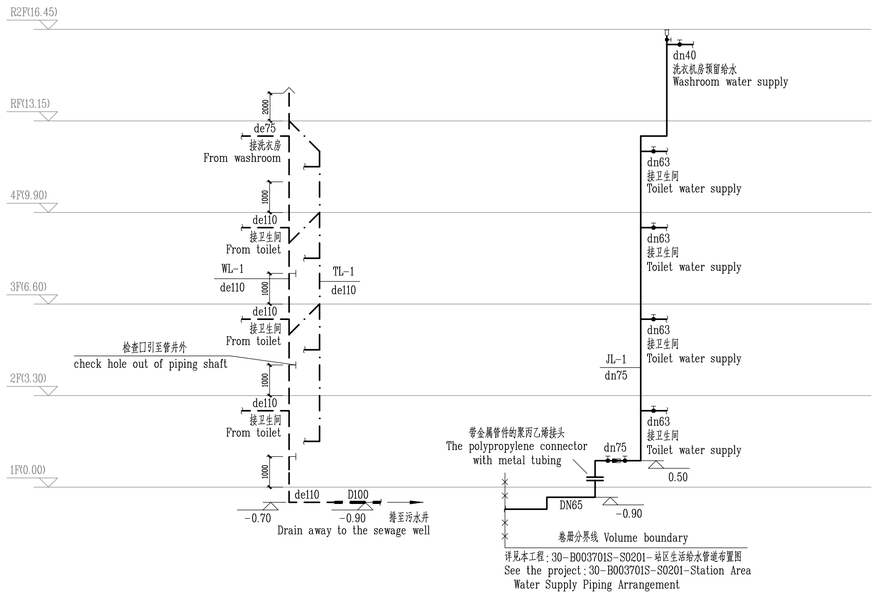
给排水管道施工安装说明

Water Supply & Drainage Construction & Installation Instruction

比例 Scale 1:100 图号 DWG NO. 30-B003701S-S0211-001 版本 REV. A

594X841





排水系统图
Drainage System Drawing

给水系统图
Water Supply System Drawing

Reference Drawings

Drwg. No.	Title

Revisions

	Name	Date	Notes

巴基斯坦默蒂亚里—拉合尔±660kV直流输电工程
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拉合尔±660kV换流站
±660kV Lahore Converter Station

设计阶段 PHASE: 施工图 Detail Design

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批准 Approved	陈世奇	2019.09.06

巴方宿舍
Dormitory Building for Pakistani
一层给排水管道布置图及系统图
The First Floor of the Water Supply & Drainage Plane Arrangement Drawing & System Drawing

比例 Scale: 1:100
图号 DWG NO.: 30-B003701S-S0211-002
版本 REV.: B

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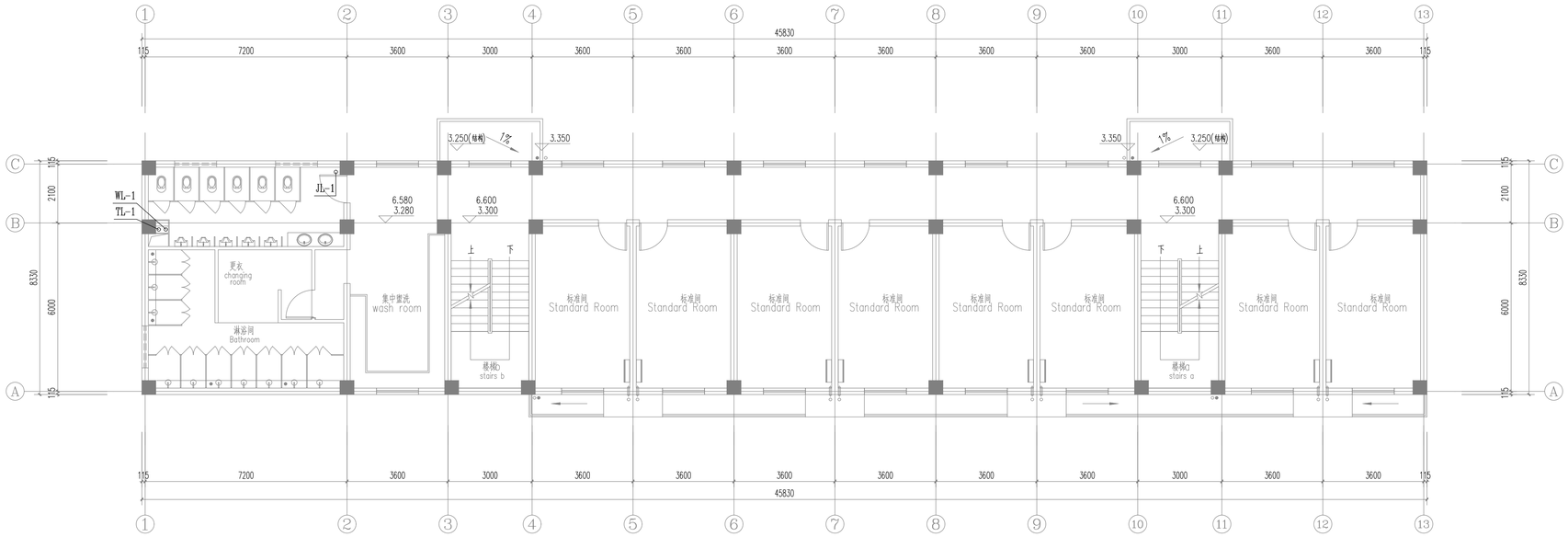

Reference Drawings

Drwg. No.	Title

Revisions

	Name	Date	Notes

巴基斯坦默蒂亚里—拉合尔±660kV直流输电工程

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拉合尔±660kV换流站

±660kV Lahore Converter Station

设计阶段 PHASE: 施工图 Detail Design

M L H V D C - L C S - E C I - D D - W F - 0 2 1 1 - 0 0 3 B

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比例 Scale	1:100	图号 DWG NO.

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Dormitory Building for Pakistani

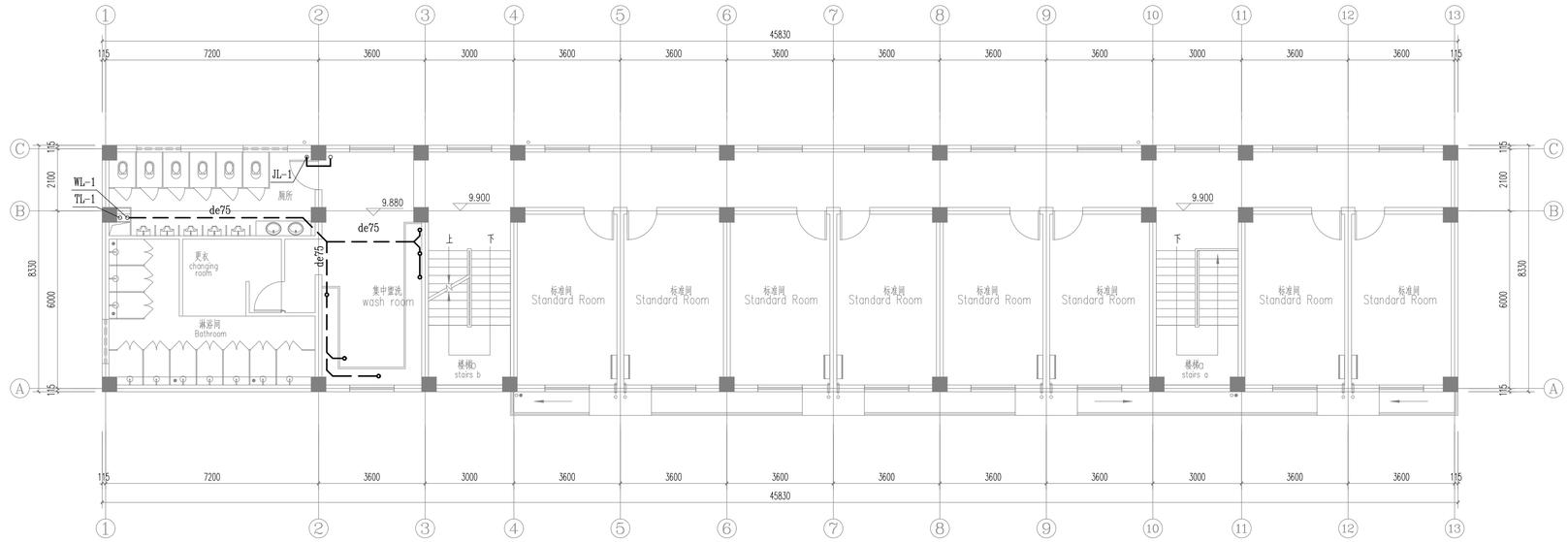
二、三层给排水管道布置图

The Second & the Third Floor of the Water Supply & Drainage Plane Arrangement Drawing

30-B003701S-S0211-003

版本 REV. B

594X841



Reference Drawings

Drwg. No.	Title

Revisions

	Name	Date	Notes

巴基斯坦默蒂亚里—拉合尔±660kV直流输电工程

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拉合尔±660kV换流站

±660kV Lahore Converter Station

设计阶段 PHASE: 施工图 Detail Design

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巴方宿舍

Dormitory Building for Pakistani

四层给排水管道布置图

The Fourth Floor of the Water Supply & Drainage Plane Arrangement Drawing

比例 Scale: 1:100

图号 DWG NO.: 30-B003701S-S0211-004

版本 REV.: B

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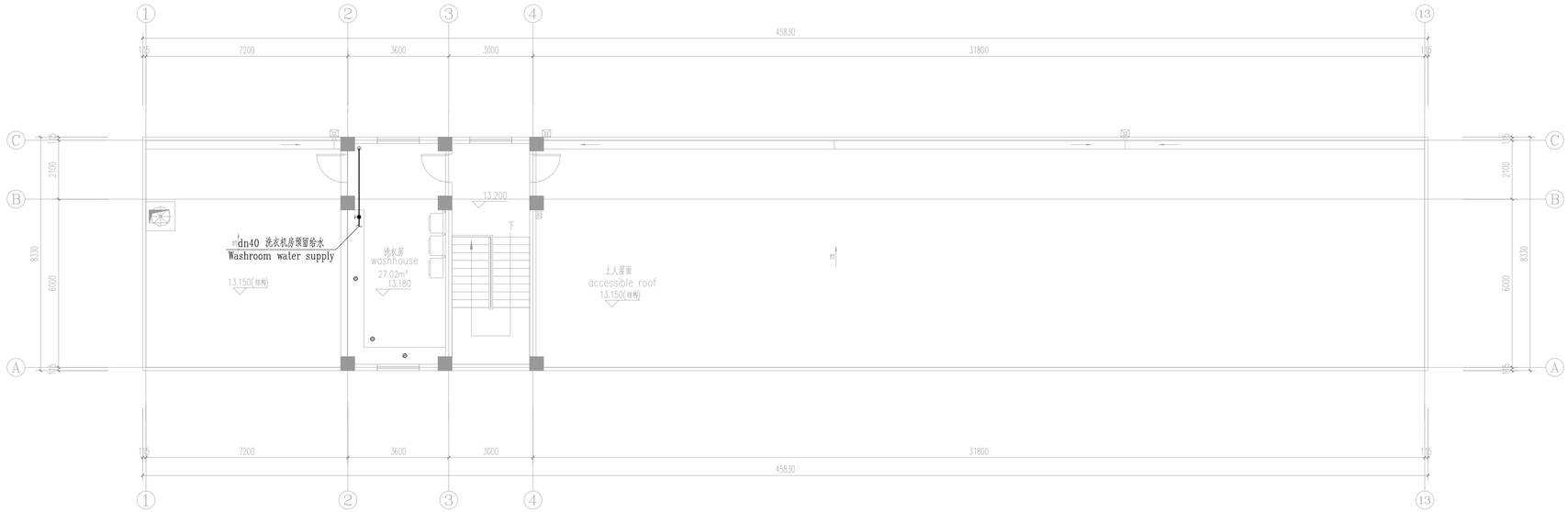

Reference Drawings

Drwg. No.	Title

Revisions

	Name	Date	Notes

巴基斯坦默蒂亚里—拉合尔±660kV直流输电工程

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拉合尔±660kV换流站

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设计阶段 PHASE: 施工图 Detail Design

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巴方宿舍

Dormitory Building for Pakistani

屋顶层给排水管道布置图

Roof Floor of the Water Supply & Drainage Plane Arrangement Drawing

比例 Scale: 1:100

图号 DWG NO.: 30-B003701S-S0211-005

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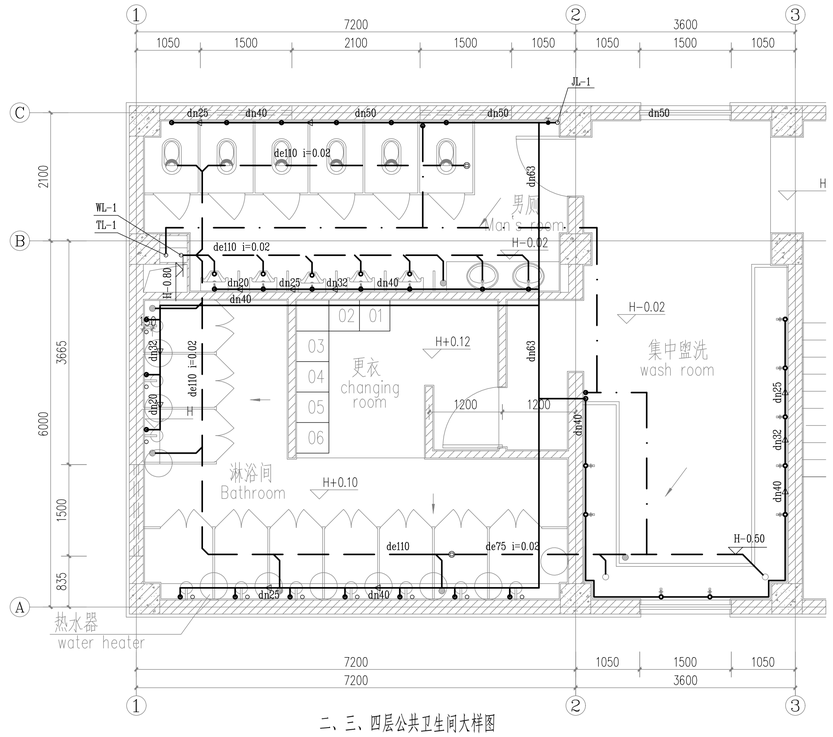
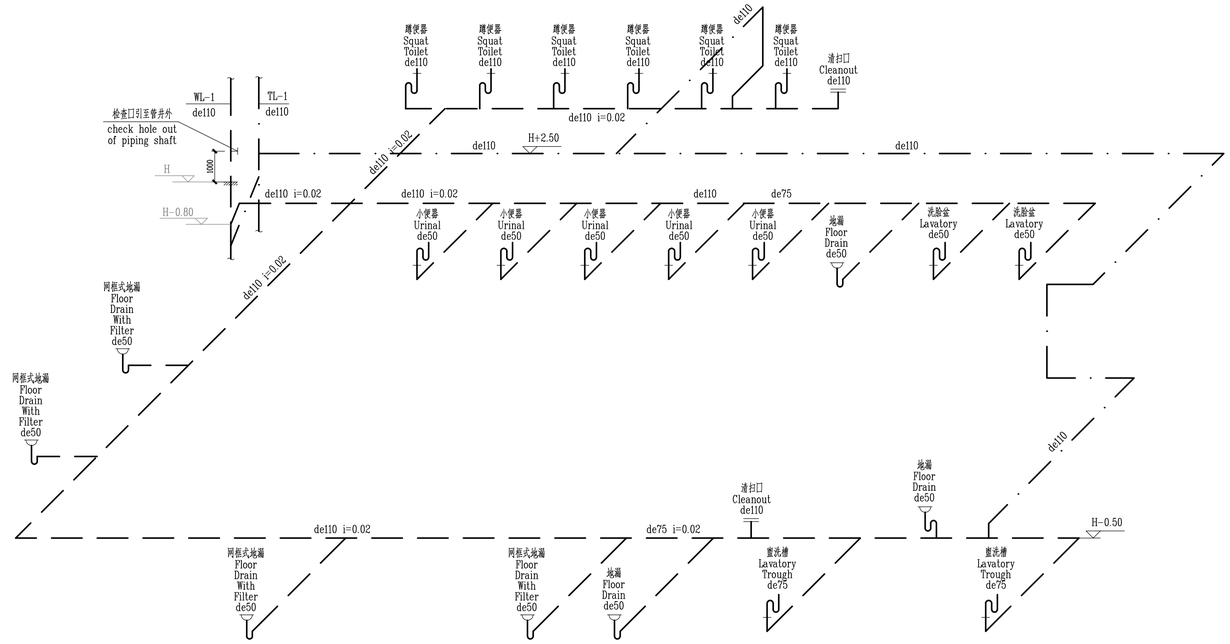
594X841



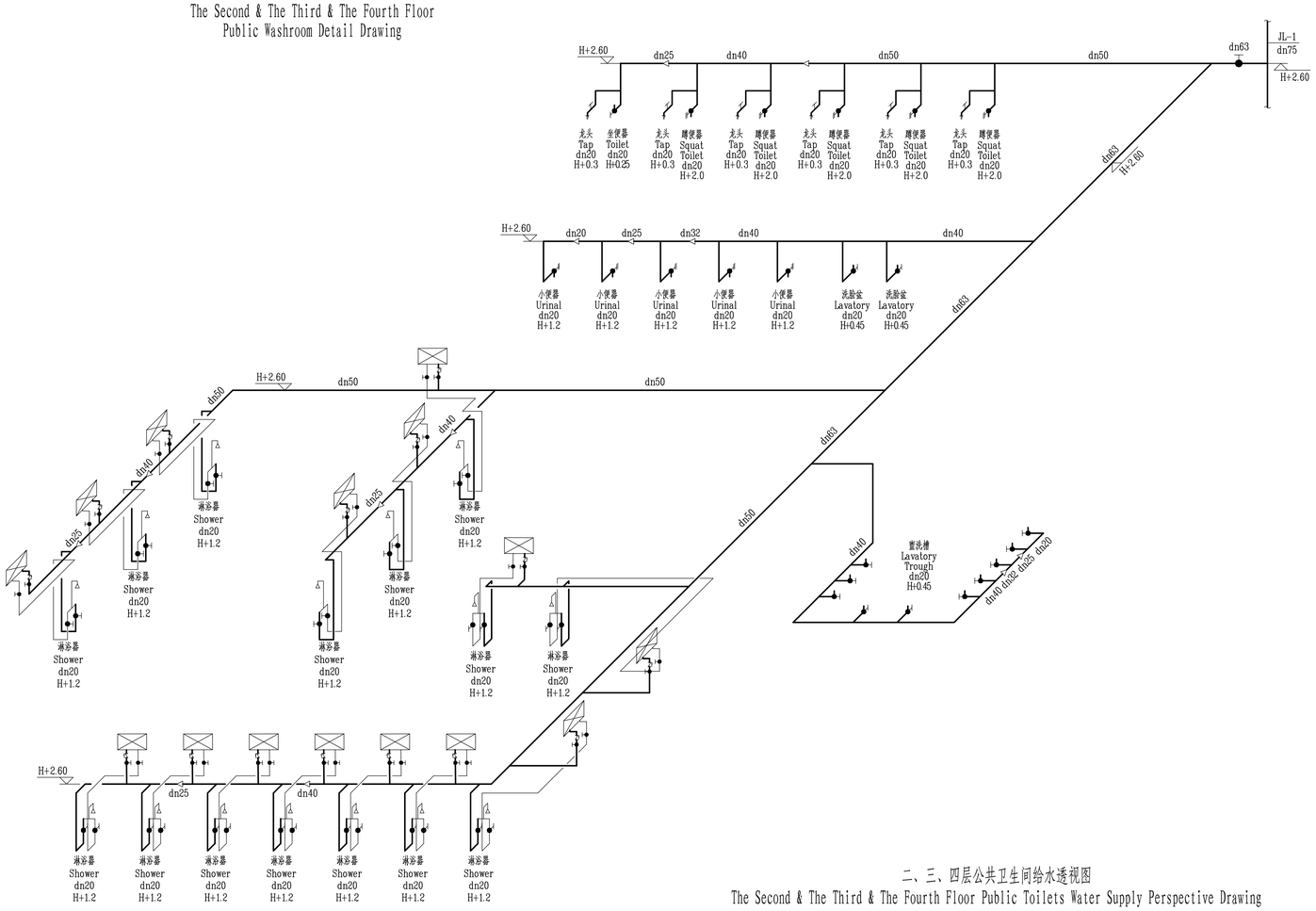
二、三、四层公共卫生间大样图

The Second & The Third & The Fourth Floor Public Washroom Detail Drawing



二、三、四层公共卫生间排水透视图

The Second & The Third & The Fourth Floor Public Toilets Drainage Perspective Drawing



二、三、四层公共卫生间给水透视图

The Second & The Third & The Fourth Floor Public Toilets Water Supply Perspective Drawing

Reference Drawings

Drwg. No.	Title

Revisions

	Name	Date	Notes

巴基斯坦默蒂亚里—拉合尔±660kV直流输电工程

±660kV HVDC Project from Matiari to Lahore in Pakistan

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拉合尔±660kV换流站

±660kV Lahore Converter Station

设计阶段 PHASE: 施工图 Detail Design

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巴方宿舍

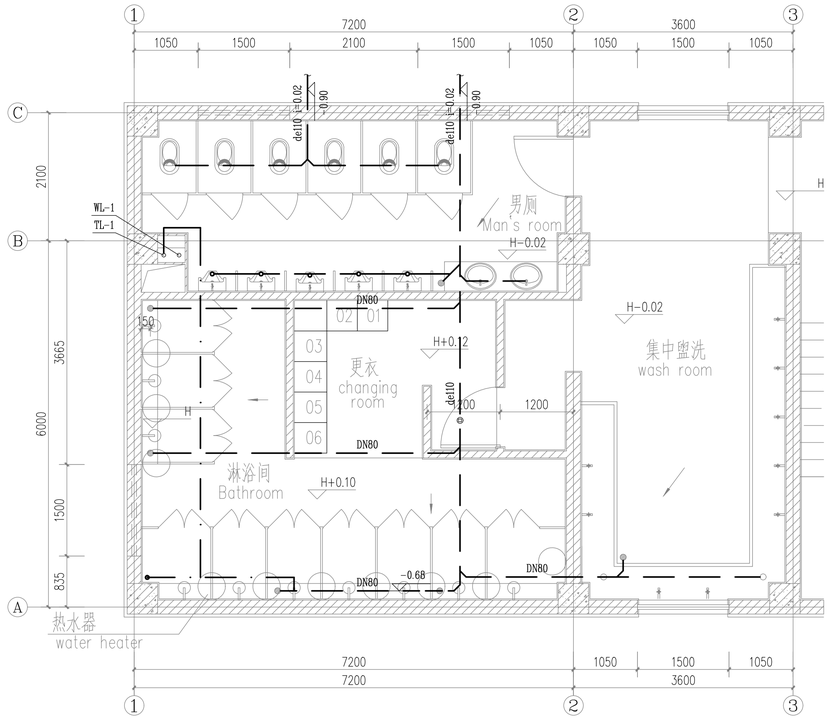
Dormitory Building for Pakistani

卫生间大样图01

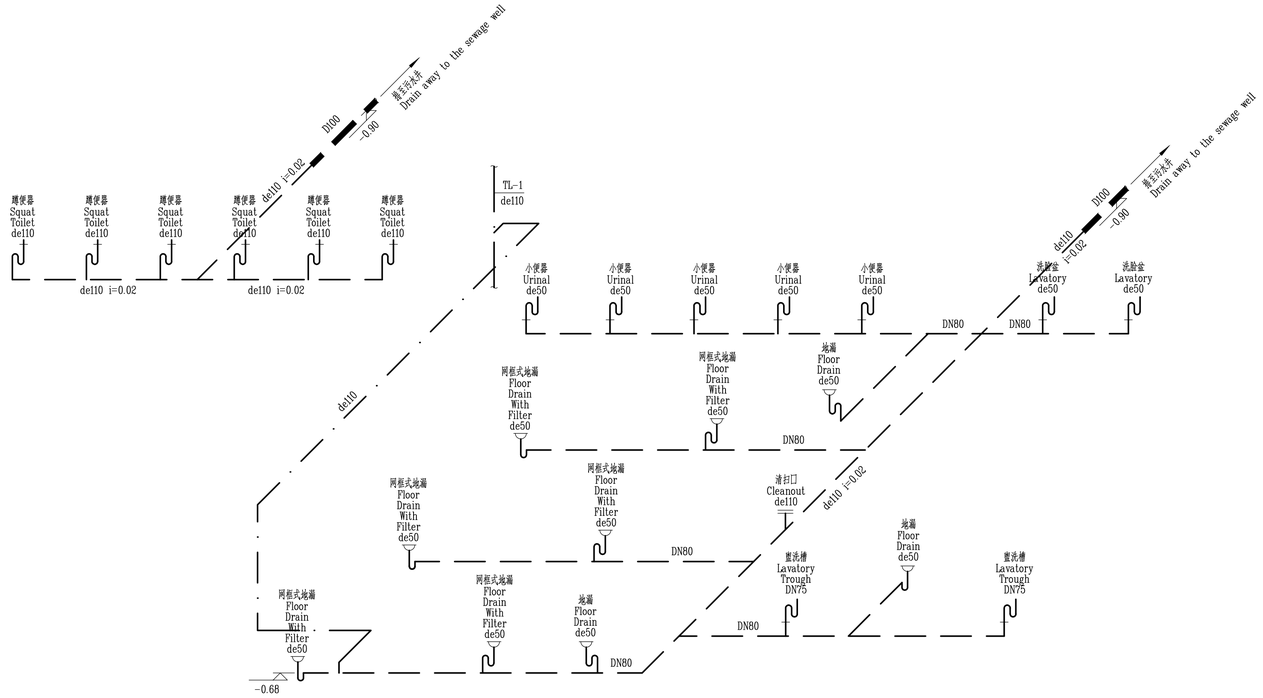
Toilet Detail Plan(01)

比例 Scale: 1:100 | 图号 DWG NO.: 30-B003701S-S0211-006 | 版本 REV.: B

594X841

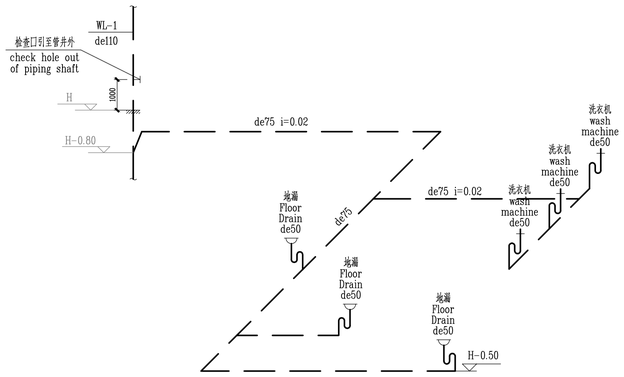


一层公共卫生间大样图
The First Floor
Public Washroom Detail Drawing



一层公共卫生间排水透视图
The First Floor Public Toilets Drainage Perspective Drawing

注：给水详见标准间大样图
P.S. For water supply, see the standard room detail drawing



洗衣房排水透视图
Washroom Drainage Perspective Drawing

Reference Drawings

Drwg. No.	Title

Revisions

	Name	Date	Notes

巴基斯坦默蒂亚里—拉合尔±660kV直流输电工程
±660kV HVDC Project from Matiari to Lahore in Pakistan

NATIONAL TRANSMISSION & DESPATCH COMPANY (NTDC) LIMITED PAKISTAN

CONTRACTOR 国家电网 State Grid 中国电力技术装备有限公司 CHINA ELECTRIC POWER EQUIPMENT AND TECHNOLOGY CO.,LTD.

SUBCONTRACTOR East China Electric Power Design Institute Co.,Ltd.

拉合尔±660kV换流站
±660kV Lahore Converter Station

设计阶段 PHASE: 施工图 Detail Design

M L H V D C - L C S - E C I - D D - W F - 0 2 1 1 - 0 0 7 B

	姓名 Name	日期 Date
设计 Designed	共复跃	2019.09.06
校核 Checked	张平	2019.09.06
审核 Reviewed	陈世奇	2019.09.06
批准 Approved	陈世奇	2019.09.06

巴方宿舍
Dormitory Building for Pakistani
卫生间大样图02
Toilet Detail Plan(02)

比例 Scale: 1:100
图号 DWG NO.: 30-B003701S-S0211-007
版本 REV.: B

594X841

编号 No.	名称 Name	规格 Parameters	单位 Unit	数量 Quan	材料 Material	单重 Single 重量(公斤)	总重 Totoal Weight(Kg)	参考图号及备注 Remark
27	井盖、盖座	Φ700	只	4				给水排水标准图集14S501-1/31、32
	Manhole cover/cover seat		Set					ARCHLIB 14S501-1/31、32
26	PP-R管与镀锌钢管过渡连接管件		只	1				
	PP-R pipe to galvanized steel pipe transition fitting		Set					
25	钢管	DN65	米	5				与室外生活水管要求一致
	Steel pipe		m					Same as outdoor water supply pipe requirements
24	清扫口	D100	只	7				
	Cleanout		Set					
23	水表	DN65,PN=1.0MPa	只	1				
	Water Meter		Set					
22	冷水PP-R管	dn75 S5	米	20	冷水PPR			含管件、接头、伸缩节、大小头、弯头、三通等配件
	Cold water PP-R		m		Cold water PP-R			Include fittings, joints, telescopic joints, reducers, elbows, and other accessories
21	冷水PP-R管	dn63 S5	米	25	冷水PPR			含管件、接头、伸缩节、大小头、弯头、三通等配件
	Cold water PP-R		m		Cold water PP-R			Include fittings, joints, telescopic joints, reducers, elbows, and other accessories
20	冷水PP-R管	dn50 S5	米	40	冷水PPR			含管件、接头、伸缩节、大小头、弯头、三通等配件
	Cold water PP-R		m		Cold water PP-R			Include fittings, joints, telescopic joints, reducers, elbows, and other accessories
19	冷水PP-R管	dn40 S5	米	100	冷水PPR			含管件、接头、伸缩节、大小头、弯头、三通等配件
	Cold water PP-R		m		Cold water PP-R			Include fittings, joints, telescopic joints, reducers, elbows, and other accessories
18	冷水PP-R管	dn32 S5	米	30	冷水PPR			含管件、接头、伸缩节、大小头、弯头、三通等配件
	Cold water PP-R		m		Cold water PP-R			Include fittings, joints, telescopic joints, reducers, elbows, and other accessories
17	冷水PP-R管	dn25 S5	米	30	冷水PPR			含管件、接头、伸缩节、大小头、弯头、三通等配件
	Cold water PP-R		m		Cold water PP-R			Include fittings, joints, telescopic joints, reducers, elbows, and other accessories
16	冷水PP-R管	dn20 S5	米	400	冷水PPR			含管件、接头、伸缩节、大小头、弯头、三通等配件
	Cold water PP-R		m		Cold water PP-R			Include fittings, joints, telescopic joints, reducers, elbows, and other accessories
15	球阀	DN15 PN=1.0MPa	只	120	冷水PPR			
	Ball Valve		Set		Cold water PP-R			
14	截止阀	DN32 PN=1.0MPa	只	1	冷水PPR			
	Ball Valve		Set		Cold water PP-R			
13	截止阀	DN40 PN=1.0MPa	只	4	冷水PPR			
	Ball Valve		Set		Cold water PP-R			
12	截止阀	DN50 PN=1.0MPa	只	4	冷水PPR			
	Ball Valve		Set		Cold water PP-R			
11	截止阀	DN65 PN=1.0MPa	只	2	冷水PPR			
	Ball Valve		Set		Cold water PP-R			
10	排气阀组(含配套球阀)	DN25 PN=1.0MPa	只	1	冷水PPR			
	Exhaust valve group (including supporting ball valve)		Set		Cold water PP-R			
9	伸顶通气帽	dn100	只	1				
	Vent cap		Set					
8	混凝土管	D100	米	20				管件、检查口、接头、弯头、顺水三通等配件均计入管长
	Concrete pipe		m					Fittings, inspection port, joint, elbow, sweep junction and other accessories are included in the pipe length
7	混凝土管	D300	米	50				管件、检查口、接头、弯头、顺水三通等配件均计入管长
	Concrete pipe		m					Fittings, inspection port, joint, elbow, sweep junction and other accessories are included in the pipe length
6	地漏	DN50	套	12				
	Floor drain		Set					
5	网框式地漏	DN50	套	24				
	Floor Drain With Filter		Set					
4	龙头	包括中所有安装附件	套	64				
	Tap	Including all components and fittings	Set					
3	小便器	包括中所有安装附件	套	20				给水排水标准图集09S304/98
	Urinal	Including all components and fittings	Set					ARCHLIB 09S304/98
2	高水箱蹲便器	包括中所有安装附件	套	24				给水排水标准图集09S304/81
	High water tank squat toilet	Including all components and fittings	Set					ARCHLIB 09S304/81
1	洗脸盆	包括中所有安装附件	套	8				给水排水标准图集09S304/37
	Lavatory	Including all components and fittings	Set					ARCHLIB 09S304/37
40								
39								
38								
37								
36								
35	止回阀	DN20 PN=1.0MPa	只	60	冷水PPR			
	check valve		Set		Cold water PP-R			
34	PVC-U排水管	de50	米	150				管件、检查口、接头、弯头、顺水三通等配件均计入管长
	PVC-U drainage pipe		m					Fittings, inspection port, joint, elbow, sweep junction and other accessories are included in the pipe length
33	PVC-U排水管	de75	米	60				管件、检查口、接头、弯头、顺水三通等配件均计入管长
	PVC-U drainage pipe		m					Fittings, inspection port, joint, elbow, sweep junction and other accessories are included in the pipe length
32	PVC-U排水管	de110	米	100				管件、检查口、接头、弯头、顺水三通等配件均计入管长
	PVC-U drainage pipe		m					Fittings, inspection port, joint, elbow, sweep junction and other accessories are included in the pipe length
31	热水PP-R管	dn50 S3.2	米	80	热水PPR			含管件、接头、伸缩节、大小头、弯头、三通等配件
	Hot water PP-R		m		Hot water PP-R			Include fittings, joints, telescopic joints, reducers, elbows, and other accessories
30	热水PP-R管	dn20 S3.2	米	40	热水PPR			含管件、接头、伸缩节、大小头、弯头、三通等配件
	Hot water PP-R		m		Hot water PP-R			Include fittings, joints, telescopic joints, reducers, elbows, and other accessories
29	电热水器	V=80L P=2Kw	只	40				暂定，以最终采购为准
	Electric water heater		Set					Tentative, subject to final purchase
28	圆形混凝土污水检查井	Φ1000	只	4				给水排水标准图集02S515/22
	Round concrete sewage inspection well		Set					ARCHLIB 02S515/22

Reference Drawings

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Revisions

	Name	Date	Notes

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M L H V D C - L C S - E C I - D D - W F - 0 2 1 1 - 0 0 8 B

	姓名 Name	日期 Date
设计 Designed	朱昊晓	2019.09.06
校核 Checked	徐平	2019.09.06
审核 Reviewed	陈北京	2019.09.06
批准 Approved	陈北京	2019.09.06

巴方宿舍

Dormitory Building for Pakistani

给排水管道材料表

Bill of Materials for Water Supply & Drainage

比例 Scale: 1:100 | 图号 DWG NO.: 30-B003701S-S0211-008 | 版本 REV.: B

420X594